A Physics-guided Hamiltonian Neural Network Framework for Reduced Magnetic Catheter Dynamics

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Abstract

This paper develops a physics-guided Hamiltonian neural network (HNN) framework for reduced magnetic catheter dynamics along the original continuum-mechanics–Hamiltonian–KAM route. A Kirchhoff–Cosserat rod with magnetic, hydrodynamic and contact interactions is first reduced by a Galerkin basis. The resulting finite-dimensional model is written as a canonical Hamiltonian system for its conservative part and as a port-Hamiltonian system when damping, fluid forces, contact losses or control inputs are present. We then define real and complex internal parameterizations of a Hamiltonian neural network and separate a fixed quadratic physical prior from the representation choice. The main experiment uses 96 COMSOL trajectories, split by complete trajectories into 64 training, 16 validation and 16 test trajectories, with five training seeds for each model. An independent explicit conservative quartic polynomial is used to compare a central fourth-order generating-function scheme, implicit midpoint, classical RK4 and two-stage fourth-order Gauss–Legendre integration. The fixed prior improves finite-time prediction under both real and complex parameterizations, while the available design does not establish a universal real/complex advantage. A corrected complex-coordinate derivation shows that complex hidden features do not by themselves represent physical dissipation. Finally, a quartic normal-form argument gives a conditional qualitative local KAM family for the explicit conservative polynomial at sufficiently small amplitudes. The conclusion is deliberately object-specific: a nonzero training loss can be interpreted as a Hamiltonian perturbation only under a common canonical chart and a suitable derivative error bound, and the KAM result does not automatically transfer to the learned or material system.

Keywords

Hamiltonian neural networks, complex-valued neural networks, magnetic flexible catheters, Galerkin reduction, port-Hamiltonian systems, symplectic integration, KAM theory

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1 Introduction

Flexible magnetic catheters are distributed mechanical systems. Their state is affected by elastic strain, inertia, magnetic actuation, fluid interaction, boundary contact and, in some settings, gravity. A spatial rod model is therefore a physically meaningful starting point (Antman 2005; Simo 1985; Abbott et al. 2020), but a model that is useful for prediction must also specify a finite-dimensional state, an energy convention, the treatment of nonconservative ports and a time-stepping rule.

The present work follows the original teacher-directed route: coupled continuum equations are reduced by a Galerkin basis, the conservative core is expressed in Hamiltonian form, a Hamiltonian neural network learns the unknown energy correction, and local invariant-torus structure is examined through KAM theory. The central scientific requirement is that each statement be attached to the object for which it is valid. The COMSOL reference trajectories represent a material and forcing configuration. The four HNNs used in the learning experiment close their explicit ports and generate a vector field from a real scalar output. The KAM calculation is performed on a separately specified analytic conservative polynomial. These objects motivate one another, but they are not identified without additional modelling and error estimates.

Two distinctions are essential. First, a fixed quadratic energy prior and a real/complex neural representation are different factors and should be compared separately. Second, a complex internal representation is not automatically a complex physical energy: in the main learning experiment the network still returns a real scalar Hamiltonian. Dissipation and control are treated through port variables. This avoids interpreting an imaginary network feature as a material loss potential, while retaining the complex-coordinate derivation that motivates the original idea.

The paper makes four contributions. (i) It supplies a mechanically grounded reduction from a coupled rod model to a finite-dimensional Hamiltonian or port-Hamiltonian system. (ii) It gives a consistent real/complex CHNN formulation, including the Wirtinger derivative factors for the coordinate convention used here. (iii) It reports a two-by-two prior/representation comparison and an independent integration study without attributing one result to the wrong factor. (iv) It states the KAM conclusion conditionally for the explicit conservative polynomial and explains what additional evidence would be needed to transfer it to a learned or material system.

2 Coupled continuum mechanics and Galerkin reduction

2.1 Kinematics and balance laws

Let $s \in [0, L]$ be the material coordinate and $r(s,t) \in \mathbb{R}^3$ the centreline. Let $d_i(s,t)$ be an orthonormal director frame. For an extensible, unshearable rod,

$$r_s = (1+\epsilon)d_3, \qquad d_{i,s} = \kappa \times d_i, \tag{1}$$

where ϵ is the axial strain and κ is the material curvature. The inextensible model is the special case $\epsilon = 0$; this is an assumption, not an identity.

Let $n(s,t)$ be the internal force, $M(s,t)$ the internal moment, $\omega(s,t)$ the angular velocity, A the cross-sectional area and J the cross-sectional inertia tensor. The local balances are

$$\begin{aligned} \rho A r_{tt} &= \partial_s n + f_{\text{mag}} + f_{\text{hydro}} + f_{\text{contact}} + f_g, \\ \rho J\dot{\omega} + \omega \times (\rho J \omega) &= \partial_s M + r_s \times n + \tau_{\text{mag}} + \tau_{\text{contact}}. \end{aligned} \tag{2}$$

The residual gravity term f_g may be negligible in a prescribed microgravity approximation, but it is not zero unless that approximation is explicitly imposed.

A compatible constitutive choice is

$$n = K_\Gamma(\Gamma - \Gamma_0), \qquad M = K_\kappa(\kappa - \kappa_0), \tag{3}$$

where Γ collects the axial and shear strain measures. Boundary conditions and boundary work have to be included before the weak form is projected.

For a point-dipole approximation with constant dipole moment μ, the magnetic interaction may be represented by

$$\begin{aligned} U_{\text{mag}} &= -\mu \cdot B(r,t), \\ f_{\text{mag}} &= \nabla_r(\mu \cdot B), \qquad \tau_{\text{mag}} = \mu \times B. \end{aligned} \tag{4}$$

A Maxwell-stress formulation is an alternative, but it must be integrated over a specified control volume rather than added to (4) without explanation. For fluid interaction it is safer to keep the force convention explicit. With relative velocity $v_r = v_f - r_t$, a typical drag contribution is proportional to $|v_r|v_r$ and an added-mass contribution depends on the relative acceleration; the precise signs depend on whether the force is written on the rod or on the fluid. In the reduced model below, these terms are therefore collected in Q_{hydro} with their implementation convention recorded in the data contract.

A unilateral contact law can be written using penetration $\delta = d_0 - d$ as

$$f_{\mathrm{contact}} = -k_c[\delta]_+^n n_c - c_c[\dot{\delta}]_+ n_c + f_{\mathrm{friction}}, \qquad [a]_+ = \max(a, 0). \tag{5}$$

Only the positive-penetration branch produces a normal contact force. The friction term is generally nonconservative.

2.2 Weak form and modal reduction

Let $\Phi(s)$ satisfy the essential boundary conditions and write

$$r(s,t) = r_0(s,t) + \Phi(s)q(t), \qquad q(t) \in \mathbb{R}^n. \tag{6}$$

Substitution into the weak form, including the boundary terms, gives a reduced equation of the form

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V(q) = Q_{\mathrm{mag}}(q,t) + Q_{\mathrm{hydro}}(q,\dot{q},t) + Q_{\mathrm{contact}}(q,\dot{q},t) + Q_g(q,t). \tag{7}$$

For example,

$$Q_{\mathrm{hydro}}(q,\dot{q},t) = \int_0^L \Phi(s)^{\mathsf{T}} f_{\mathrm{hydro}}(s,t)\,\mathrm{d}s, \tag{8}$$

and analogous projections define the magnetic, contact and gravity forces. The conservative potential may contain elastic energy and those magnetic or contact contributions that are gradients of scalar potentials. Velocity-dependent drag, dry friction and externally prescribed actuation remain in the nonconservative part.

The theoretical reduction permits a configuration-dependent mass matrix. The supplied learning experiment uses a four-mode basis and a fixed stored matrix M_4, so its canonical momentum is

$$p = M_4\dot{q}. \tag{9}$$

This specialization is important: replacing M_4 by a learned or configuration-dependent mass metric would change the canonical normalization, the prior and the numerical experiment.

3 Hamiltonian and port-Hamiltonian formulation

3.1 Legendre transform and canonical equations

For the general reduced conservative part define

$$p = M(q)\dot{q}, \qquad H(q,p) = \frac{1}{2}p^{\mathsf{T}}M(q)^{-1}p + V(q). \tag{10}$$

Let $z = (q,p) \in \mathbb{R}^{2n}$ and

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \qquad \Omega = \sum_{j=1}^{n} \mathrm{d}q_j \wedge \mathrm{d}p_j. \tag{11}$$

The closed conservative system is

$$\dot{z} = J\nabla H(z), \qquad \dot{q} = \nabla_p H, \qquad \dot{p} = -\nabla_q H. \tag{12}$$

If $N(q) = M(q)^{-1}$ and $H = \frac{1}{2}p^{\mathsf{T}}N(q)p + V(q)$, then

$$\dot{q} = N(q)p, \qquad \dot{p}_i = -\frac{1}{2}p^{\mathsf{T}}N_{,q_i}(q)p - V_{,q_i}(q). \tag{13}$$

The first term in $\dot{p}_i$ is the geometric contribution of the kinetic metric. It disappears only when the inverse mass matrix is constant. Likewise, a nonlinear potential requires its true gradient; $K(q)q$ is only a special linear or quadratic case.

3.2 Nonconservative forces as ports

Let Q_{nc} collect drag, contact loss, gravity residuals and control. A structure-compatible representation is

$$\dot{z} = (J - R)\nabla H(z) + Gu, \qquad y = G^{\mathsf{T}}\nabla H(z), \tag{14}$$

where $R = R^{\mathsf{T}} \succeq 0$. It follows that

$$\dot{H} = -\nabla H^{\mathsf{T}} R \nabla H + u^{\mathsf{T}}y + \partial_t H. \tag{15}$$

This is the appropriate place for physical dissipation. Standard conservative KAM theory applies to the closed Hamiltonian part, not directly to a system with $R \neq 0$ or an arbitrary time-dependent input.

3.3 Four-mode learning coordinates

For the fixed four-mode matrix, let U satisfy, up to stored numerical precision,

$$U^{\mathsf{T}}M_4U = I, \qquad U^{\mathsf{T}}K_4U = \Lambda^2. \tag{16}$$

Define

$$S = \mathrm{diag}(S_q, S_p), \qquad S_q = U\Lambda^{-1/2}, \qquad S_p = M_4 U \Lambda^{1/2}, \tag{17}$$

so that $S^{\mathsf{T}} J S = J$. The learning state is $y = (Q, P) = S(q, p)$. The fixed quadratic prior is

$$H_2(Q, P) = \frac{1}{2} P^{\mathsf{T}} M_{\mathrm{norm}}^{-1} P + \frac{1}{2} Q^{\mathsf{T}} K_{\mathrm{norm}} Q. \tag{18}$$

The stored matrices and the coordinate map are part of the experimental definition; the map does not nondimensionalize time or identify the network energy with the complete material energy.

4 Complex coordinates and Hamiltonian neural networks

4.1 Complex canonical coordinates

For paired canonical coordinates $y = (Q, P)$ define

$$\xi = \frac{Q + iP}{2}, \qquad \bar{\xi} = \frac{Q - iP}{2}. \tag{19}$$

Then $Q = \xi + \bar{\xi}$ and $P = -i(\xi - \bar{\xi})$. The canonical Poisson bracket is

$$\{F, G\} = \nabla_Q F^{\mathsf{T}} \nabla_P G - \nabla_P F^{\mathsf{T}} \nabla_Q G, \tag{20}$$

and the coordinate convention gives

$$\{\xi_j, \bar{\xi}_k\} = -\frac{i}{2} \delta_{jk}. \tag{21}$$

For a real Hamiltonian $H(Q, P)$, Wirtinger differentiation yields

$$\dot{\xi} = -\frac{i}{2} \frac{\partial H}{\partial \bar{\xi}}, \qquad \dot{\bar{\xi}} = \frac{i}{2} \frac{\partial H}{\partial \xi}. \tag{22}$$

The factors in (22) depend on the definition in (19); changing the factor in the coordinate definition changes the bracket and the evolution coefficient.

A formal complexification may be written as $\mathcal{H} = H + iG$. Under the convention

$$\dot{\xi} = -\frac{i}{2} \frac{\partial \mathcal{H}}{\partial \bar{\xi}}, \tag{23}$$

the additional term is $\frac{1}{2} \partial_{\bar{\xi}} G$. This term is not automatically a physical damping force. To represent a prescribed port, G must satisfy the corresponding integrability conditions in the chosen coordinates. For this reason the main experiment uses a real output Hamiltonian and keeps physical nonconservative effects in (14); the complex variables are used to describe an internal representation and to derive the conservative complex-coordinate equations.

4.2 Real and complex CHNNs

The no-prior model learns a scalar energy $H_\theta(y)$ and generates

$$\dot{y}_\theta = J \nabla_y H_\theta(y). \tag{24}$$

The gray-box model adds the fixed prior,

$$H_\theta^{\mathrm{prior}}(y) = H_2(y) + H_{\mathrm{res},\theta}(y). \tag{25}$$

The real network uses real affine maps and real hidden features. The complex network uses $Wz + b$, separate real and imaginary hyperbolic tangents,

$$\sigma(u + iv) = \tanh(u) + i \tanh(v), \tag{26}$$

and a real-part readout. Its learned object is still a real scalar energy. Split complex activations need not satisfy Cauchy–Riemann equations; holomorphicity is required only if a holomorphic model is explicitly intended.

The four factorial models are R (real/no prior), GR (real/gray-box), C (complex/no prior) and GC (complex/gray-box). A residual MLP is a separate descriptive baseline. The prior effects are $R - GR$ and $C - GC$, representation effects are $R - C$ and $GR - GC$, and the interaction is

$$(GR - GC) - (R - C). \tag{27}$$

4.3 Training loss and dynamical perturbation

Let $D_{\mathrm{train}} = \mathrm{diag}(d_1, \ldots, d_8)$ be fixed from the training data. For a trajectory with 801 saved states, the stored rollout metric is

$$X_{msi} = \frac{1}{800} \sum_{k=1}^{800} \frac{1}{8} \left\| D_{\mathrm{train}}^{-1} (y_{bmsi,k} - y_{i,k}) \right\|_2^2. \tag{28}$$

This metric is a finite-time average and is not a uniform vector-field norm.

Suppose, conditionally, that a learned Hamiltonian and a target Hamiltonian are defined in the same canonical chart and with the same energy gauge. Let $\Delta H_\theta = H_\theta - H_*$. Then

$$\dot{y}_\theta - \dot{y}_* = J \nabla \Delta H_\theta. \tag{29}$$

A sufficient local perturbation statement requires a derivative bound such as

$$\sup_{y \in \mathcal{U}} \|\nabla \Delta H_\theta(y)\| \le \delta. \tag{30}$$

A nonzero rollout loss, by itself, does not establish (30). This is the precise bridge between the statistical learning problem and the dynamical KAM problem.

5 Structure-preserving time integration

5.1 Independent conservative comparison

To separate numerical structure from learning, define the explicit real polynomial

$$H_{\mathrm{dec}}(W) = \frac{1}{2}\sum_{j=1}^{4}\omega_j(X_j^2+Y_j^2) + \sum_{a\le b\le c} c_{abc}W_aW_bW_c + \sum_{a\le b\le c\le d} c_{abcd}W_aW_bW_cW_d, \qquad W=(X,Y). \tag{31}$$

The stored decimal coefficient file defines this polynomial. It is an independent conservative object and is not asserted to equal the full material Hamiltonian.

5.2 Central generating-function scheme

For an autonomous real Hamiltonian, let $u^+ = \Phi_h(u^-)$ be the exact endpoints and define

$$m = \frac{1}{2}(u^+ + u^-), \qquad \delta = u^+ - u^-. \tag{32}$$

Locally, $\delta = F(m,h)$. With $A = D_mF$, exact symplecticity gives

$$(I+A/2)^{\mathsf{T}}J(I+A/2) - (I-A/2)^{\mathsf{T}}J(I-A/2) = A^{\mathsf{T}}J + JA = 0. \tag{33}$$

Thus $-JF$ is locally a gradient. A generating function S_h satisfies

$$\nabla_m S_h = -JF, \qquad F = J\nabla_m S_h. \tag{34}$$

The existing central fourth-order construction expands S_h through order four and solves the resulting implicit endpoint equations by Newton iteration. An exact implicit root of a generating-function map can be symplectic without conserving the original energy exactly; finite solver tolerances do not prove exact symplecticity of a program.

The comparison uses central fourth order, implicit midpoint, classical RK4 and two-stage fourth-order Gauss–Legendre, all applied to the same H_{dec}, initial conditions, reference solution and cost accounting.

6 Data, experimental protocol and results

6.1 Trajectory split and evaluation

The supplied reduced-catheter data contain 96 COMSOL trajectories. Complete trajectories, rather than individual time samples, are split into 64 training, 16 validation and 16 test trajectories. Each trajectory has 801 states over 0–0.2 s with $\Delta t = 2.5\times10^{-4}$ s. The 16 test trajectories belong to six orbit groups. Five independent training seeds are retained for each model, and the epoch-300 checkpoint is used without selecting on test performance.

The original descriptive metric takes the median over five seeds for each model–trajectory pair and then summarizes the 16 test values. The resulting values are shown in Table 1.

Table 1. Descriptive rollout metric for the five models.

Model	Mean	Median
R	1.66533×10^{3}	1.37061×10^{3}
GR	1.26587×10^{1}	9.57033×10^{0}
C	1.66528×10^{3}	1.37065×10^{3}
GC	1.26640×10^{1}	9.60739×10^{0}
Residual MLP	2.20953×10^{9}	2.20953×10^{9}

The fixed prior reduces the finite-test rollout metric under both parameterizations. This is evidence for the prior in the stated finite design, not evidence that the prior recovers every term of the material energy.

6.2 Factorial effects and sensitivity

The five effects are summarized in Table 2. Positive values in the first two rows favour including the prior; positive values in the next two rows favour the complex implementation under the stated metric.

Table 2. Factorial effects and available sensitivity intervals.

Effect	Estimate	Original 95% interval	New adjusted interval
$R-GR$	1652.68	$[1043.01, 2314.78]$	$[332.78, 2778.00]$
$C-GC$	1652.62	$[1037.70, 2311.88]$	$[332.58, 2777.95]$
$R-C$	0.05316	$[-0.08880, 0.17862]$	$[-0.11404, 0.98582]$
$GR-GC$	-0.00532	$[-0.14163, 0.12850]$	$[-0.01274, -0.00103]$
Interaction	-0.05848	$[-0.17323, 0.06401]$	$[-0.99095, 0.10964]$

Both prior effects retain their improvement direction under the prescribed weighting, aggregation and leave-one checks. The sealed analysis does not resolve an additional complex benefit. The post-experiment group/seed sensitivity calculation gives a negative gray-box real-minus-complex contrast, favouring GR in this design, while the no-prior representation contrast and the interaction remain unresolved. The intervals rely on the stated finite-design resampling assumptions and do not establish population coverage or universal representation superiority.

6.3 Integration results

For the same H_{dec}, the state error and energy change are measured by

$$e_{\mathrm{state}} = \max_t \frac{\|W_{\mathrm{num}}(t) - W_{\mathrm{ref}}(t)\|_2}{\|W_0\|_2}, \qquad e_H = \max_n \frac{|H_{\mathrm{dec}}(W_n) - H_{\mathrm{dec}}(W_0)|}{H_{2,\mathrm{dec}}(W_0)} \tag{35}$$

The tested regime is weakly nonlinear and the selected amplitude is a solver scale, not a KAM radius. All 80 tested configurations completed. On the fixed step-size grid, midpoint did not attain the three prescribed state-error targets. The central fourth-order method had smaller observed energy changes than RK4, but its integration cost was approximately 2.59–4.02 times that of RK4 at the selected state-accuracy targets. Gauss was faster than the central method at 10^{-4} and 10^{-5}, while the central method was faster on the selected 10^{-3} configurations. These results establish an energy–cost trade-off on one polynomial, not uniform performance superiority.

7 Normal form and conditional local KAM structure

7.1 Normal form of the explicit polynomial

The KAM object is H_{dec} in (31). Write

$$H_{\mathrm{dec}} = H_2 + H_3 + H_4, \tag{36}$$

where H_j is homogeneous of degree j. In complex canonical variables, the quadratic part is

$$H_2 = \sum_{j=1}^{4} \omega_j I_j, \qquad I_j = \xi_j \bar{\xi}_j, \tag{37}$$

up to the fixed scaling convention. Under the nonresonance assumptions needed to solve the cubic homological equation, choose a cubic generator χ_3 such that

$$\{\chi_3, H_2\} = -H_3. \tag{38}$$

The fourth-order normal form has the structure

$$h_{\mathrm{NF}}(I) = \omega \cdot I + \frac{1}{2} I^{\mathsf{T}} T I, \tag{39}$$

where T is computed from H_4 and the cubic correction. The nondegeneracy condition is

$$\det T \neq 0. \tag{40}$$

The full analytic polynomial is not replaced by its truncation. If C is the finite analytic canonical transformation, then

$$H_{\mathrm{dec}} \circ C = h_{\mathrm{NF}}(I) + R_{\mathrm{Lie}}(I, \theta), \tag{41}$$

where the remainder contains all terms not removed by the finite transformation.

7.2 KAM conclusion

Let the frequency vector satisfy the Diophantine condition

$$|k \cdot \omega(I)| \geq \gamma |k|^{-\tau}, \qquad k \in \mathbb{Z}^4 \setminus \{0\}, \qquad \tau > 3, \tag{42}$$

and suppose the analytic norm of R_{Lie} is sufficiently small on a common complex neighbourhood. Classical nondegenerate KAM theory then gives the following object-specific statement.

Proposition 1. *For the explicitly specified analytic polynomial H_{dec}, if the quartic normal form satisfies $\det T \neq 0$ and the analytic remainder and Diophantine hypotheses of a nondegenerate KAM theorem hold, then a positive-measure family of analytic invariant four-tori exists at sufficiently small amplitudes near the elliptic equilibrium. The proposition specifies neither a numerical amplitude threshold nor a particular torus, and it does not certify the learned predictor or the full material system.*

The quadratic prior alone has zero twist. Its linear trajectories can be written in action–angle variables, but the nondegenerate KAM persistence argument requires the nonlinear frequency map and the full analytic remainder. Observed closed curves, small energy drift or a stable power spectrum are numerical diagnostics rather than a proof of KAM persistence.

7.3 Why transfer to the learned system is conditional

For a learned Hamiltonian H_θ, transfer would require at least: (i) a common canonical coordinate chart; (ii) a consistent energy gauge; (iii) analyticity or a suitable finite-regularity theorem in the neighbourhood of interest; (iv) a bound such as (30) or a stronger analytic norm bound; and (v) preservation of the nondegenerate frequency map. A finite trajectory loss does not supply these quantities. For a damped or forced port-Hamiltonian system, standard conservative KAM theory is not the applicable theorem without a separate dissipative or forced invariant-manifold analysis.

8 Discussion and modelling guidance

The continuum derivation shows why the Hamiltonian core is natural, but it also identifies which terms must remain outside a conservative potential. The Galerkin step explains the origin of the reduced coordinates and makes the fixed four-mode mass matrix an explicit experimental specialization rather than an unstated physical identity.

The prior and representation effects should be selected in separate comparisons. A lower error for GR or GC establishes that the prescribed quadratic prior helps the stated finite-time prediction task. It does not prove complete material identification. The real/complex comparison is likewise a comparison of the implemented widths, activations, optimization and data protocol. The results do not support a universal ordering of real and complex neural networks.

The complex Hamiltonian derivation remains useful for the original research direction: it gives a compact coordinate language, exposes the canonical bracket and provides a route to normal-form calculations. Its physical interpretation must remain controlled. A formal imaginary component G is not a damping law until the generated vector field is shown to match the desired port and power balance. In the present experiment, the safe and reproducible choice is a real scalar energy with explicit port variables for nonconservative effects.

The independent polynomial study should also remain separate from the learned field. The central generating-function scheme is an existing structure-preserving method; its observed energy–cost trade-off is a numerical result for H_{dec}. Applying the same method to H_θ would require a new cost and error comparison because automatic-differentiation and implicit-solve costs depend on the learned network.

Table 3. Scope of the main conclusions.

Object	Supported conclusion and remaining requirement
Reduced material model	Hamiltonian core plus port terms after Galerkin reduction; requires boundary, force and truncation specifications.
Learned H_θ	Finite-design prediction comparison; KAM transfer requires common coordinates and analytic/derivative error control.
Explicit H_{dec}	Conditional local KAM family under normal-form, twist, Diophantine and small-remainder assumptions.
Discrete integrator	State, energy and cost trade-offs on H_{dec}; RK4 is not generally exact symplectic.

9 Conclusion

We have rewritten the CHNN–Hamiltonian–KAM route as a single mechanically grounded workflow for reduced magnetic catheter dynamics. The coupled rod equations provide the continuum origin, Galerkin projection gives the finite-dimensional model, and the Legendre transform produces the Hamiltonian core. Damping, hydrodynamic forces, contact losses and control are retained through port-Hamiltonian terms. Real and complex CHNNs are then compared under a factorial prior/representation design, while an independent conservative polynomial is used for integration and local KAM analysis.

In the supplied finite design, the fixed quadratic prior improves prediction under both parameterizations. The available results do not establish a universal real/complex advantage. The central fourth-order scheme trades smaller observed energy changes for higher cost than RK4 at selected targets, with no consistent advantage over Gauss. The explicit polynomial admits a qualitative local KAM family under the stated analytic, twist and Diophantine assumptions. That conclusion is intentionally not transferred to the learned or material system without a common canonical chart and an appropriate derivative or analytic error estimate.

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Author contributions

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Not applicable to this computational study.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The coefficient files, analysis scripts, source-index records and complete comparison tables should accompany the submission package. The final statement will be updated to match the actual repository and access conditions.

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